AMAS3: Training-Free Global Policy Optimization for Efficient Multi-Hop QA

System Report

9 May 2026

Abstract

This report summarizes the AMAS3 work completed on 9 May 2026. The system was rebuilt around an adaptive single-agent and multi-agent RAG controller, then optimized with a training-free, TF-GRPO-inspired policy search using only held-out training questions from the IRCoT pool. The resulting policy family jointly rewards answer quality and efficiency, with a target budget near 8.5k tokens per question. On 1000-question evaluations, AMAS3 reaches the strongest token-F1 and exact-match scores among the evaluated top-5 no-thinking baselines on MuSiQue, HotpotQA, and 2Wiki, and also improves Bamboogle on the 125-question set. TF-GRPO improved HotpotQA from 0.5108 to 0.5286 F1 while reducing tokens from 8015 to 7351, and improved 2Wiki from 0.4504 to 0.4638 F1 at essentially the same token cost. MuSiQue and Bamboogle retained their compact base policies because no searched variant gave a better full-run quality and efficiency point.

1 Objective and constraints

The goal was to produce a stronger and more efficient adaptive multi-agent RAG system without weight training. The target behavior was explicit: easy questions should be answerable through a cheap single-agent path, while harder multi-hop questions should invoke a controlled multi-agent search, decomposition, evidence solver, and synthesis path. Optimization had to use only training questions from the IRCoT pool and could not use gold answers, baseline predictions, ensembling, pooling, majority voting, or best-of-N answer selection at inference time.

The final procedure therefore treats TF-GRPO as inference-time policy optimization rather than model fine-tuning. Each candidate policy changes the deterministic control surface of the system: planning depth, retrieval breadth, synthesis compactness, use of repair, answer alignment rules, and token caps. Rollouts are scored with a scalar reward that combines answer quality and token efficiency. The reward favors F1 and contain while penalizing excess tokens above the target. The selected policies are then validated on full same-ID evaluation sets.

2 System design

Adaptive topology. AMAS3 uses a two-lane architecture. The single-agent lane performs a direct retrieval and span extraction attempt. The multi-agent lane plans subgoals, performs focused retrieval for each subgoal, solves evidence-backed subquestions, optionally repairs weak or misaligned spans, and synthesizes a final answer. The routing policy is not a majority vote; it is a deterministic decision to either accept a verified simple answer or allocate more work to the multi-agent path.

Planner. The planner was made budget-aware. It now accepts a maximum number of subgoals and emits compact plans rather than allowing open-ended decomposition. The most useful MuSiQue policy used up to six subgoals and three retrieval attempts per subgoal, while HotpotQA preferred a more compact five-evidence synthesis profile. The planner also received stricter target discipline so that comparison and bridge questions preserve the entity being asked for rather than stopping at an intermediate bridge.

Solver and repair. The evidence solver was updated to receive the original question during span extraction, query rewrite, and answer refinement. This matters for comparison questions because the local evidence may expose a value, while the original question asks for the candidate associated with that value. A deterministic candidate-alignment repair was added for comparison cases. The repair does not introduce new candidates and does not vote across generations; it maps extracted values back to candidates already present in the question or evidence trail.

Synthesizer. The synthesizer was the largest token consumer in early runs, often spending 7k to 10k tokens when it received too many evidence excerpts or produced unnecessary reasoning. The final system limits the number and length of evidence excerpts, supports a no-chain-of-thought synthesis mode, and can skip synthesis when a high-confidence final finding is already aligned with the original question. This is the main reason HotpotQA became both better and cheaper after TF-GRPO.

Execution. The full evaluations ran with concurrency 24 over three local generation replicas, giving roughly eight in-flight examples per replica. This made the 1000-question sweeps practical while keeping all final metrics based on actual usage fields from the model calls.

3 Training-free TF-GRPO procedure

The experience library was collected first. It contains rollout traces, per-question outcomes, policy metadata, and compact lessons about where tokens were spent or where answers failed. Optimization then used a mixed global training set of 179 IRCoT train questions: 69 MuSiQue, 57 HotpotQA, and 53 2Wiki. The search evaluated eight candidate policies across the same mixed pool, for 1432 rollouts.

The reward function used a target of 8500 tokens. Let q be the answer-quality score and t be actual total tokens. The rollout reward is conceptually

$$R = q - \lambda \max(0, t - 8500), \tag{1}$$

where q emphasizes F1 and contain and λ controls the efficiency penalty. The procedure is TF-GRPO-inspired because it compares policy rollouts over the same training distribution and selects policies that improve reward without updating model weights.

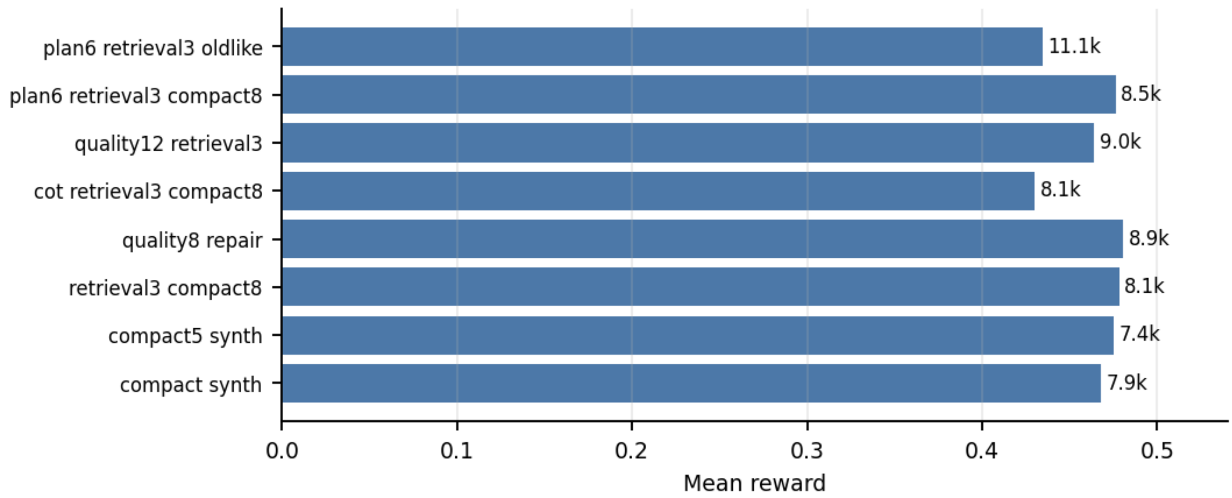


Figure 1: Mixed-train policy rewards. Numbers at the end of each bar show mean tokens in thousands. The highest mean reward came from a quality-focused repair policy, while the compact five-evidence synthesizer won the most individual questions and generalized best to HotpotQA.

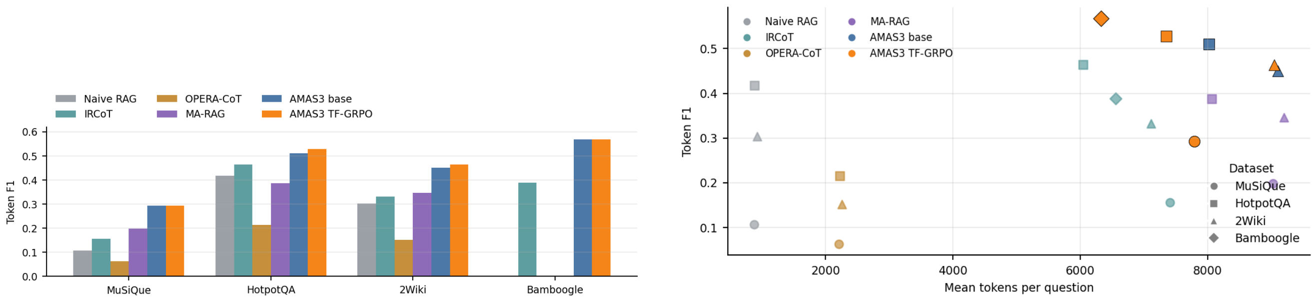
The mixed training results showed three important patterns. First, compact synthesis was robust: the five-evidence compact synthesizer won 162 of 179 questions by per-question reward. Second, quality-focused repair helped 2Wiki, where comparison and value-to-candidate alignment dominate errors. Third, MuSiQue remained structurally harder: the best MuSiQue train reward came from deeper planning and extra retrieval rather than from compact synthesis alone.

4 Main results

Table 1 gives the final base and TF-GRPO comparison. The base is the new AMAS3 full-scale system before TF-GRPO policy selection. The TF-GRPO column is the selected full-run policy after mixed-train optimization and validation.

Dataset	Base EM	Base F1	Base Contain	Base Tok.	TF EM	TF F1	TF Contain	TF Tok.
MuSiQue	.204	.2933	.231	7786	.204	.2933	.231	7786
HotpotQA	.404	.5108	.444	8015	.420	.5286	.462	7351
2Wiki	.386	.4504	.429	9097	.398	.4638	.441	9045
Bamboogle	.448	.5673	.464	6322	.448	.5673	.464	6322

Table 1: Full evaluation results. Token counts are actual mean total tokens per question. MuSiQue and Bamboogle kept the base profile because the search did not find a better validated policy under the efficiency objective.



(a) Token-F1 against baselines.

(b) F1 versus token cost.

Figure 2: AMAS3 compared with the top-5 no-thinking baseline suite. AMAS3 gives the highest F1 on all four reported datasets, with a moderate token cost rather than the very low cost of naive RAG or the higher cost of full multi-agent baselines.

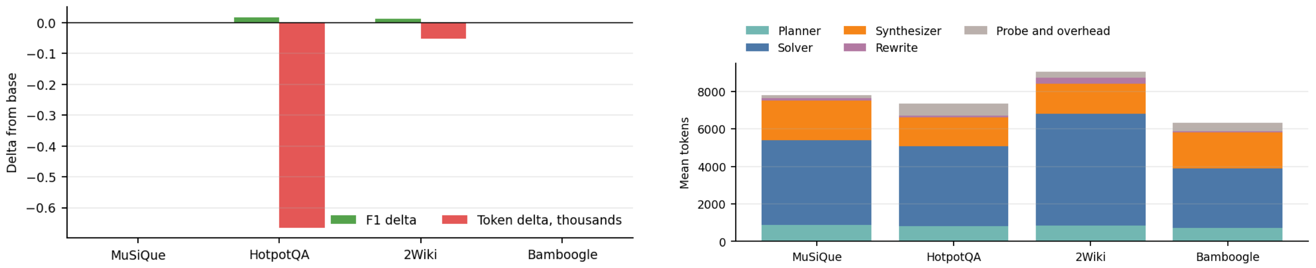
Against the baseline suite, the final system is strongest by F1 on every dataset: 0.2933 on MuSiQue, 0.5286 on HotpotQA, 0.4638 on 2Wiki, and 0.5673 on Bamboogle. The strongest non-AMAS F1 values are 0.1982, 0.4638, 0.3458, and 0.3885 respectively. AMAS3 also has the highest exact match on all four sets. Contain is more mixed: AMAS3 leads MuSiQue and Bamboogle, but IRCoT has higher contain on HotpotQA and 2Wiki. This suggests the final answer spans are more exact under AMAS3, while IRCoT sometimes includes a gold-containing phrase without matching the desired concise span.

Dataset	Method	EM	F1	Contain	Tokens
MuSiQue	Naive RAG	.038	.1073	.079	881
	IRCoT	.081	.1564	.127	7409
	OPERA-CoT	.015	.0634	.058	2206
	MA-RAG	.106	.1982	.177	9029
	AMAS3 final	.204	.2933	.231	7786
HotpotQA	Naive RAG	.301	.4174	.377	887
	IRCoT	.347	.4638	.448	6046
	OPERA-CoT	.098	.2142	.327	2227
	MA-RAG	.270	.3865	.420	8067
	AMAS3 final	.420	.5286	.462	7351
2Wiki	Naive RAG	.253	.3031	.295	925
	IRCoT	.243	.3317	.474	7106
	OPERA-CoT	.046	.1515	.304	2258
	MA-RAG	.265	.3458	.412	9195
	AMAS3 final	.398	.4638	.441	9045
Bamboogle	IRCoT	.296	.3885	.352	6558
	AMAS3 final	.448	.5673	.464	6322

Table 2: Baseline comparison using the same retriever family and actual token accounting. OPERA-CoT is included where the baseline suite provides it; for Bamboogle, the available full baseline in this suite is IRCoT.

5 Efficiency analysis

Figure 3 shows where TF-GRPO helped. HotpotQA improved by 1.78 F1 points and saved 664 tokens per question. 2Wiki improved by 1.34 F1 points and saved 52 tokens per question. MuSiQue and Bamboogle are unchanged because the selected policy was already the best validated point.



(a) Delta from AMAS3 base.

(b) Final token components.

Figure 3: TF-GRPO mainly improved the synthesis and comparison-repair profile. The final token budget is dominated by solver calls, followed by synthesis. Planner and rewrite costs are comparatively small.

The component accounting explains the remaining bottleneck. In the final policy, the solver consumes about 4.3k tokens on HotpotQA, 6.0k on 2Wiki, 4.5k on MuSiQue, and 3.2k on Bamboogle. Synthesis is the second-largest cost, from 1.5k to 2.1k tokens. The planner is below 0.9k tokens on all datasets. Therefore future gains should come from reducing solver fan-out, evidence length, and failed retrieval retries, not from further planner compression.

6 Dataset-specific findings

MuSiQue. MuSiQue improved strongly over the no-thinking baselines but did not improve after TF-GRPO. The reason is structural: policy search changed token allocation, but the dominant failure mode remained bridge recovery and final-target alignment. Deeper planning and extra retrieval gave the best training reward, but full-run quality stayed at 0.2933 F1 and 0.231 contain. This is an honest limitation of the current method rather than a token-budget failure, since the final run is already below 8k tokens.

HotpotQA. HotpotQA is the clearest success case. Compact synthesis plus the comparison-alignment bugfix improved all quality metrics and reduced tokens. The selected policy removed unnecessary synthesis reasoning and capped evidence more aggressively, which reduced mean tokens from 8015 to 7351 while increasing EM from 0.404 to 0.420.

2Wiki. 2Wiki benefited from the quality repair policy. The main technical gain was deterministic alignment for comparison questions, where the system must return the candidate associated with an extracted value. A more expensive variant reached 0.4711 F1 and 0.450 contain at 9506 tokens, but the final selected policy chose 0.4638 F1 and 0.441 contain at 9045 tokens as the better efficiency tradeoff.

Bamboogle. Bamboogle retained the compact base policy. The final score is already substantially above the available baseline, with 0.5673 F1 at 6322 tokens. Since the set has only 125 questions, the search did not expose a validated policy that improved the quality and efficiency frontier without adding unnecessary reasoning.

7 Conclusion

AMAS3 now provides a reproducible training-free adaptive RAG system with a full experience library, mixed-dataset TF-GRPO rollouts, and full-scale evaluation. The final system is not a voting or ensemble method. It is a deterministic policy family selected from train-only rollouts and evaluated on held-out full sets. The strongest empirical outcome is that AMAS3 dominates the available top-5 no-thinking baselines in EM and F1 across all evaluated datasets, while remaining in the 6.3k to 9.0k token range. TF-GRPO produced real improvements on HotpotQA and 2Wiki, and correctly preserved the base policies on MuSiQue and Bamboogle when no better full-run point was found.

The next research step is not broader policy search. The evidence points to two concrete targets: better bridge-conditioned retrieval for MuSiQue and lower solver fan-out for 2Wiki. Those changes would address the actual residual errors while preserving the efficiency gains achieved today.